

B.E. I year (E&I)
Class Test – III (December' 2023)
2AMRC1: Applied Mathematics-II

Note: Attempt only four questions.
Time: 70 min

Maximum Marks: 20

- Q.1 States and prove Baye's Theorem. (5)
- Q.2 (i) During war, 1 ship out of 9 was sunk on an average in making a certain voyage. What was the probability that exactly 3 out of a convoy of 6 ships would arrive safely? (5)
(ii) If on an average one ship in every ten is wrecked, find the probability that out of 5 ships expected to arrive, 4 at least will arrive safely.
- Q.3 Solve the Equation $X^4 - 2X^3 - 21X^2 + 22X + 40 = 0$ whose roots are in arithmetic Progression (5)
- Q.4 Solve by Cardan's Method $X^3 - 3X^2 + 12X + 16 = 0$ (5)
- Q.5 Explain With Examples: a) Union of Fuzzy Set b) Truth Value of Fuzzy Set (5)